# IGCSE 0607 Practice Questions — E1.8

## Standard form

**Total Questions:** 9 — **Total Marks:** 48 Marks

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**1** **(a)** Write 450000 in standard form.

..... **[1]**

**(b)** Write $3.08 \times 10^{-4}$ as an ordinary number.

..... **[1]**

**(c)** Work out.

$(2 \times 10^3) \times (4 \times 10^5)(0)$

*Give your answer in standard form.*

..... **[1]**

**2** **(a)** Write 452000 in standard form.

..... **[1]**

**(b)** Write $6.07 \times 10^{-3}$ as an ordinary number.

..... **[1]**

**(c)** Work out $(4 \times 10^4) \times (3 \times 10^5)$.
Give your answer in standard form.

..... **[2]**

**3** Consider the following numerical values.

**(a)** Convert the following numbers as instructed.

**(i)** Write 0.00034 in standard form.

..... **[1]**

**(ii)** Write $5.2 \times 10^4$ as an ordinary number.

..... **[1]**

**(b)** Work out, giving your answer in standard form:

$(2 \times 10^3) \times (4 \times 10^5)$

..... **[2]**

**4** The mass of the Earth is $5.97 \times 10^{24}$ kg.
The mass of the Moon is $7.35 \times 10^{22}$ kg.

**(a)** Calculate the difference between the mass of the Earth and the mass of the Moon.
Give your answer in standard form.

..... *kg* **[2]**

**(b)** The mass of the Sun is $1.99 \times 10^{30}$ kg.
Find the value of $k$ when

mass of the Sun = k ×mass of the Earth
Give your answer to 3 significant figures.

$k =$ ..... **[3]**

**5** **(a)** Work out the exact value of

$$\frac{5.4 \times 10^5 + 6 \times 10^4}{2 \times 10^{-3}}$$

Give your answer in standard form.

..... **[3]**

**(b)** Evaluate $\sqrt{2.5 \times 10^9}$.
Give your answer in standard form.

..... **[2]**

**(c)** Given that $p = 2 \times 10^x$ and $q = 8 \times 10^y$, find an expression for $p^2 \div q$.
Give your answer in standard form in terms of $x$ and $y$.

..... **[3]**

**6** The mass of the Sun is $1.99 \times 10^{30}$ kg.
The mass of Jupiter is $1.90 \times 10^{27}$ kg.

**(a)** Calculate how many times greater the mass of the Sun is than the mass of Jupiter.
Give your answer in standard form correct to 3 significant figures.

..... **[3]**

**(b)** A comet has a mass of $4.5 \times 10^{13}$ kg.
Calculate the mass of the comet in grams.
Give your answer in standard form.

..... $g$ **[2]**

**7** **(a)** Calculate, giving your answer in standard form.

$3.4 \times 10^7 + 2.1 \times 10^6$

..... **[2]**

**(b)** A computer can do a calculation in $4 \times 10^{-9}$ seconds.
Work out how many of these calculations the computer can do in 5 minutes.
Give your answer in standard form.

..... **[3]**

**8** **(a)** The mass of Earth is $5.97 \times 10^{24}$ kg and the mass of the Moon is $7.35 \times 10^{22}$ kg.
Calculate the difference between the mass of Earth and the mass of the Moon.
Give your answer in standard form.

..... kg **[2]**

**(b)** The distance from the Sun to Earth is $1.50 \times 10^{8}$ km.
The speed of light is $3.00 \times 10^{5}$ km/s.

Calculate the time it takes for light to travel from the Sun to Earth.
Give your answer in standard form.

..... $s$ **[2]**

**(c)** The Earth is modelled as a sphere with radius $r = 6.37 \times 10^{6}$ m.
Calculate the surface area of the Earth.
Give your answer in standard form, correct to 3 significant figures.
The surface area, $A$, of a sphere with radius $r$ is $A = 4\pi r^2$.

..... $m^2$ **[3]**

**9** A large cylindrical water tank has a base radius of $4.5 \times 10^2$ cm and a height of $1.2 \times 10^3$ cm.

**(a)** Calculate the volume of the water tank in $cm^3$.
Give your answer in standard form correct to 4 significant figures.

..... $cm^3$ **[3]**

**(b)** The empty tank is being filled with water at a constant rate of $1.5 \times 10^{-4}$ $m^3$ per second.
Note that $1\ m^3 = 1 \times 10^6\ cm^3$.

Calculate the time taken to fill the tank completely.
Give your answer in hours, written in standard form correct to 3 significant figures.

..... hours **[4]**